

继电保护考题回忆

选择题，不定项选择

1. 两相星形接法多用于 (B) 电路 做宜
A 变压器保护回路 B 低压馈线回路 C 特高压输电线路 D 想不起来了。。。
2. 下列保护可用于远后备保护的是 (AB)
A 距离保护 B 过电流保护 C 纵联保护 D 电流差动保护
3. 在距离保护中，为什么距离III段通常无需震荡闭锁元件 A.
A 靠自身延时即可躲过振荡闭锁
B 距离III段通常采用故障分量原理制成的元件
C 和 D 想不起来了
4. 两相短路接地故障中共有 (C) 个故障回路
A 1 B 2 C 3 D 6
- 5.

填空题

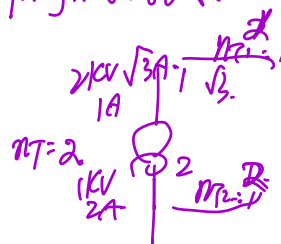
1. 一道依据返回系数分别计算动作电流和返回电流的填空，很简单
2. 阶段式距离保护通常由 () 组成，其中 () 组成主保护， () 构成后备保护
电流速断. 限时电流速断. 定时限过电流.
3. 变压器差动保护中，若变压器为Yd11接法，电流互感器的配置原则是 ()，两侧电流互感器变比与变压器变比之间关系是 ()
主保护 后备保护
星形侧采用Δ接线. 三角形侧采用Y接线.
4. 整定计算通常是在保护最 () 的情况下进行的；灵敏度校验通常是在保护最 () 的情况下进行的。
 $\frac{n_{T2}}{n_{T1}} = \frac{n}{\sqrt{3}}$
不校验.
5. 双母线固定元件电流差动保护中，若将I母线的负荷全部转移到II母线上，那么大差动回路中出现的电流为 (D)，I母线差动回路中出现的电流为 (D)。
D: 母线负荷电流
6. 自动重合闸的重合闸成功率和重合闸设施的 () 有关，重合闸正确动作率和重合闸设备的 () 有关。
重合闸? 选择性?
7. 电力系统发生不对称单相接地故障时，故障点的零序电压 () 零序功率方向元件 () 电压死区
较高 不拒动
8. 故障启动发讯闭锁式方向高频保护，若在故障时线路上存在高频信号，那么这个信号在信号性质上属于 () 信号，且通常在线路的 () 位置。
闭锁 两端线路
9. 双侧电源送电线路两侧重合闸配合方式为：一侧仅投入 () 装置，另一侧投入 () 装置。
同步检知 无压检知
10. 发电机失磁后，若处于临界失步状态，发电机机端阻抗圆呈现为 ()。
等无功阻抗圆
11. 一般来说，远后备保护的 () 最高。
选择性.
12. 若过电流保护灵敏度不足，可通过引入 () 电流分量来解决。
零序.
13. 若在闭锁方向式纵联保护中发生非全相运行，应采用 () 侧电压作为测量电压

$$K_{re} = \frac{I_{re}}{I_{op}}$$

电流速断. 限时电流速断. 定时限过电流.

主保护 后备保护
星形侧采用Δ接线. 三角形侧采用Y接线.

$$\frac{n_{T2}}{n_{T1}} = \frac{n}{\sqrt{3}}$$



例题① 振荡三相对称无零序故障
性略有

简答题

1.振荡闭锁状态下和故障状态下电气测量量有哪些区别?

书上已经总结了

2.就电流差动保护来说,引起变压器、母线和线路的不平衡电流的因素各有那些异同?

母线应从区外故障时故障所在励磁支路过饱和,全部非故障侧支路电流作为不平衡电流的角度作答;

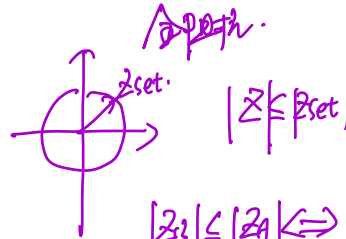
变压器书中已总结;

线路书中已总结

3.在复平面画图说明全阻抗继电器的比相式动作特性(使用电流和电压相量表示);并简述相位比较原理和幅值比较原理的互换方法

4.

- ② 振荡电气量周期变化, 阻抗继电器
- ③ 阻抗元件一周期的动作二次, 阻抗不动作。

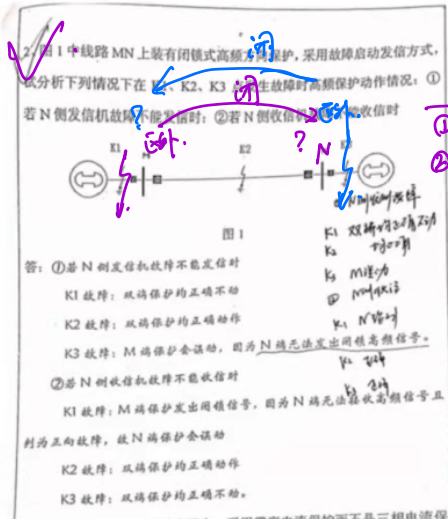


$$|Z| \leq |Z_{set}| \Leftrightarrow \arg \frac{Z}{Z_{set}} \in (-90^\circ, 90^\circ)$$

$$Z_D = Z_C - Z_D$$

$$Z_A = Z_C + Z_D$$

$$\therefore \arg \frac{Z + Z_{set}}{Z_{set} - Z} \in (-90^\circ, 90^\circ)$$



答: ①若N侧发信机故障不能发信时
K1故障: 双端保护均正确不动
K2故障: 双端保护均正确动作
K3故障: M端保护会误动, 因为N端无法发出闭锁信号。
②若N侧收信机故障不能收信时
K1故障: M端保护发出闭锁信号, 因为N端无法接收闭锁信号且判为正向故障, 故N端保护会误动
K2故障: 双端保护均正确动作
K3故障: 双端保护均正确不动。

5.

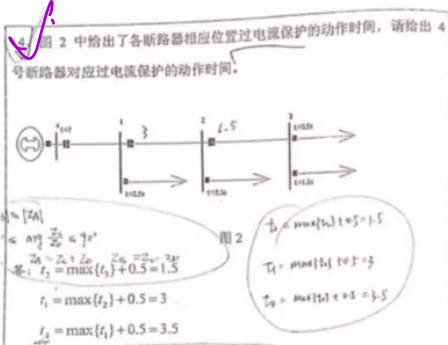


图2中给出了各断路器相应位置过电流保护的动作时间, 请给出4号断路器对应过电流保护的動作时间。
 $t_1 = \max\{t_2\} + 0.5 = 1.5$
 $t_2 = \max\{t_3\} + 0.5 = 1.5$
 $t_3 = \max\{t_4\} + 0.5 = 1.5$
 $t_4 = \max\{t_5\} + 0.5 = 1.5$
 $t_5 = \max\{t_6\} + 0.5 = 1.5$
 $t_6 = \max\{t_7\} + 0.5 = 1.5$
 $t_7 = \max\{t_8\} + 0.5 = 1.5$
 $t_8 = \max\{t_9\} + 0.5 = 1.5$
 $t_9 = \max\{t_{10}\} + 0.5 = 1.5$
 $t_{10} = \max\{t_{11}\} + 0.5 = 1.5$
 $t_{11} = \max\{t_{12}\} + 0.5 = 1.5$
 $t_{12} = \max\{t_{13}\} + 0.5 = 1.5$
 $t_{13} = \max\{t_{14}\} + 0.5 = 1.5$
 $t_{14} = \max\{t_{15}\} + 0.5 = 1.5$
 $t_{15} = \max\{t_{16}\} + 0.5 = 1.5$
 $t_{16} = \max\{t_{17}\} + 0.5 = 1.5$
 $t_{17} = \max\{t_{18}\} + 0.5 = 1.5$
 $t_{18} = \max\{t_{19}\} + 0.5 = 1.5$
 $t_{19} = \max\{t_{20}\} + 0.5 = 1.5$
 $t_{20} = \max\{t_{21}\} + 0.5 = 1.5$
 $t_{21} = \max\{t_{22}\} + 0.5 = 1.5$
 $t_{22} = \max\{t_{23}\} + 0.5 = 1.5$
 $t_{23} = \max\{t_{24}\} + 0.5 = 1.5$
 $t_{24} = \max\{t_{25}\} + 0.5 = 1.5$
 $t_{25} = \max\{t_{26}\} + 0.5 = 1.5$
 $t_{26} = \max\{t_{27}\} + 0.5 = 1.5$
 $t_{27} = \max\{t_{28}\} + 0.5 = 1.5$
 $t_{28} = \max\{t_{29}\} + 0.5 = 1.5$
 $t_{29} = \max\{t_{30}\} + 0.5 = 1.5$
 $t_{30} = \max\{t_{31}\} + 0.5 = 1.5$
 $t_{31} = \max\{t_{32}\} + 0.5 = 1.5$
 $t_{32} = \max\{t_{33}\} + 0.5 = 1.5$
 $t_{33} = \max\{t_{34}\} + 0.5 = 1.5$
 $t_{34} = \max\{t_{35}\} + 0.5 = 1.5$
 $t_{35} = \max\{t_{36}\} + 0.5 = 1.5$
 $t_{36} = \max\{t_{37}\} + 0.5 = 1.5$
 $t_{37} = \max\{t_{38}\} + 0.5 = 1.5$
 $t_{38} = \max\{t_{39}\} + 0.5 = 1.5$
 $t_{39} = \max\{t_{40}\} + 0.5 = 1.5$
 $t_{40} = \max\{t_{41}\} + 0.5 = 1.5$
 $t_{41} = \max\{t_{42}\} + 0.5 = 1.5$
 $t_{42} = \max\{t_{43}\} + 0.5 = 1.5$
 $t_{43} = \max\{t_{44}\} + 0.5 = 1.5$
 $t_{44} = \max\{t_{45}\} + 0.5 = 1.5$
 $t_{45} = \max\{t_{46}\} + 0.5 = 1.5$
 $t_{46} = \max\{t_{47}\} + 0.5 = 1.5$
 $t_{47} = \max\{t_{48}\} + 0.5 = 1.5$
 $t_{48} = \max\{t_{49}\} + 0.5 = 1.5$
 $t_{49} = \max\{t_{50}\} + 0.5 = 1.5$
 $t_{50} = \max\{t_{51}\} + 0.5 = 1.5$
 $t_{51} = \max\{t_{52}\} + 0.5 = 1.5$
 $t_{52} = \max\{t_{53}\} + 0.5 = 1.5$
 $t_{53} = \max\{t_{54}\} + 0.5 = 1.5$
 $t_{54} = \max\{t_{55}\} + 0.5 = 1.5$
 $t_{55} = \max\{t_{56}\} + 0.5 = 1.5$
 $t_{56} = \max\{t_{57}\} + 0.5 = 1.5$
 $t_{57} = \max\{t_{58}\} + 0.5 = 1.5$
 $t_{58} = \max\{t_{59}\} + 0.5 = 1.5$
 $t_{59} = \max\{t_{60}\} + 0.5 = 1.5$
 $t_{60} = \max\{t_{61}\} + 0.5 = 1.5$
 $t_{61} = \max\{t_{62}\} + 0.5 = 1.5$
 $t_{62} = \max\{t_{63}\} + 0.5 = 1.5$
 $t_{63} = \max\{t_{64}\} + 0.5 = 1.5$
 $t_{64} = \max\{t_{65}\} + 0.5 = 1.5$
 $t_{65} = \max\{t_{66}\} + 0.5 = 1.5$
 $t_{66} = \max\{t_{67}\} + 0.5 = 1.5$
 $t_{67} = \max\{t_{68}\} + 0.5 = 1.5$
 $t_{68} = \max\{t_{69}\} + 0.5 = 1.5$
 $t_{69} = \max\{t_{70}\} + 0.5 = 1.5$
 $t_{70} = \max\{t_{71}\} + 0.5 = 1.5$
 $t_{71} = \max\{t_{72}\} + 0.5 = 1.5$
 $t_{72} = \max\{t_{73}\} + 0.5 = 1.5$
 $t_{73} = \max\{t_{74}\} + 0.5 = 1.5$
 $t_{74} = \max\{t_{75}\} + 0.5 = 1.5$
 $t_{75} = \max\{t_{76}\} + 0.5 = 1.5$
 $t_{76} = \max\{t_{77}\} + 0.5 = 1.5$
 $t_{77} = \max\{t_{78}\} + 0.5 = 1.5$
 $t_{78} = \max\{t_{79}\} + 0.5 = 1.5$
 $t_{79} = \max\{t_{80}\} + 0.5 = 1.5$
 $t_{80} = \max\{t_{81}\} + 0.5 = 1.5$
 $t_{81} = \max\{t_{82}\} + 0.5 = 1.5$
 $t_{82} = \max\{t_{83}\} + 0.5 = 1.5$
 $t_{83} = \max\{t_{84}\} + 0.5 = 1.5$
 $t_{84} = \max\{t_{85}\} + 0.5 = 1.5$
 $t_{85} = \max\{t_{86}\} + 0.5 = 1.5$
 $t_{86} = \max\{t_{87}\} + 0.5 = 1.5$
 $t_{87} = \max\{t_{88}\} + 0.5 = 1.5$
 $t_{88} = \max\{t_{89}\} + 0.5 = 1.5$
 $t_{89} = \max\{t_{90}\} + 0.5 = 1.5$
 $t_{90} = \max\{t_{91}\} + 0.5 = 1.5$
 $t_{91} = \max\{t_{92}\} + 0.5 = 1.5$
 $t_{92} = \max\{t_{93}\} + 0.5 = 1.5$
 $t_{93} = \max\{t_{94}\} + 0.5 = 1.5$
 $t_{94} = \max\{t_{95}\} + 0.5 = 1.5$
 $t_{95} = \max\{t_{96}\} + 0.5 = 1.5$
 $t_{96} = \max\{t_{97}\} + 0.5 = 1.5$
 $t_{97} = \max\{t_{98}\} + 0.5 = 1.5$
 $t_{98} = \max\{t_{99}\} + 0.5 = 1.5$
 $t_{99} = \max\{t_{100}\} + 0.5 = 1.5$
 $t_{100} = \max\{t_{101}\} + 0.5 = 1.5$
 $t_{101} = \max\{t_{102}\} + 0.5 = 1.5$
 $t_{102} = \max\{t_{103}\} + 0.5 = 1.5$
 $t_{103} = \max\{t_{104}\} + 0.5 = 1.5$
 $t_{104} = \max\{t_{105}\} + 0.5 = 1.5$
 $t_{105} = \max\{t_{106}\} + 0.5 = 1.5$
 $t_{106} = \max\{t_{107}\} + 0.5 = 1.5$
 $t_{107} = \max\{t_{108}\} + 0.5 = 1.5$
 $t_{108} = \max\{t_{109}\} + 0.5 = 1.5$
 $t_{109} = \max\{t_{110}\} + 0.5 = 1.5$
 $t_{110} = \max\{t_{111}\} + 0.5 = 1.5$
 $t_{111} = \max\{t_{112}\} + 0.5 = 1.5$
 $t_{112} = \max\{t_{113}\} + 0.5 = 1.5$
 $t_{113} = \max\{t_{114}\} + 0.5 = 1.5$
 $t_{114} = \max\{t_{115}\} + 0.5 = 1.5$
 $t_{115} = \max\{t_{116}\} + 0.5 = 1.5$
 $t_{116} = \max\{t_{117}\} + 0.5 = 1.5$
 $t_{117} = \max\{t_{118}\} + 0.5 = 1.5$
 $t_{118} = \max\{t_{119}\} + 0.5 = 1.5$
 $t_{119} = \max\{t_{120}\} + 0.5 = 1.5$
 $t_{120} = \max\{t_{121}\} + 0.5 = 1.5$
 $t_{121} = \max\{t_{122}\} + 0.5 = 1.5$
 $t_{122} = \max\{t_{123}\} + 0.5 = 1.5$
 $t_{123} = \max\{t_{124}\} + 0.5 = 1.5$
 $t_{124} = \max\{t_{125}\} + 0.5 = 1.5$
 $t_{125} = \max\{t_{126}\} + 0.5 = 1.5$
 $t_{126} = \max\{t_{127}\} + 0.5 = 1.5$
 $t_{127} = \max\{t_{128}\} + 0.5 = 1.5$
 $t_{128} = \max\{t_{129}\} + 0.5 = 1.5$
 $t_{129} = \max\{t_{130}\} + 0.5 = 1.5$
 $t_{130} = \max\{t_{131}\} + 0.5 = 1.5$
 $t_{131} = \max\{t_{132}\} + 0.5 = 1.5$
 $t_{132} = \max\{t_{133}\} + 0.5 = 1.5$
 $t_{133} = \max\{t_{134}\} + 0.5 = 1.5$
 $t_{134} = \max\{t_{135}\} + 0.5 = 1.5$
 $t_{135} = \max\{t_{136}\} + 0.5 = 1.5$
 $t_{136} = \max\{t_{137}\} + 0.5 = 1.5$
 $t_{137} = \max\{t_{138}\} + 0.5 = 1.5$
 $t_{138} = \max\{t_{139}\} + 0.5 = 1.5$
 $t_{139} = \max\{t_{140}\} + 0.5 = 1.5$
 $t_{140} = \max\{t_{141}\} + 0.5 = 1.5$
 $t_{141} = \max\{t_{142}\} + 0.5 = 1.5$
 $t_{142} = \max\{t_{143}\} + 0.5 = 1.5$
 $t_{143} = \max\{t_{144}\} + 0.5 = 1.5$
 $t_{144} = \max\{t_{145}\} + 0.5 = 1.5$
 $t_{145} = \max\{t_{146}\} + 0.5 = 1.5$
 $t_{146} = \max\{t_{147}\} + 0.5 = 1.5$
 $t_{147} = \max\{t_{148}\} + 0.5 = 1.5$
 $t_{148} = \max\{t_{149}\} + 0.5 = 1.5$
 $t_{149} = \max\{t_{150}\} + 0.5 = 1.5$
 $t_{150} = \max\{t_{151}\} + 0.5 = 1.5$
 $t_{151} = \max\{t_{152}\} + 0.5 = 1.5$
 $t_{152} = \max\{t_{153}\} + 0.5 = 1.5$
 $t_{153} = \max\{t_{154}\} + 0.5 = 1.5$
 $t_{154} = \max\{t_{155}\} + 0.5 = 1.5$
 $t_{155} = \max\{t_{156}\} + 0.5 = 1.5$
 $t_{156} = \max\{t_{157}\} + 0.5 = 1.5$
 $t_{157} = \max\{t_{158}\} + 0.5 = 1.5$
 $t_{158} = \max\{t_{159}\} + 0.5 = 1.5$
 $t_{159} = \max\{t_{160}\} + 0.5 = 1.5$
 $t_{160} = \max\{t_{161}\} + 0.5 = 1.5$
 $t_{161} = \max\{t_{162}\} + 0.5 = 1.5$
 $t_{162} = \max\{t_{163}\} + 0.5 = 1.5$
 $t_{163} = \max\{t_{164}\} + 0.5 = 1.5$
 $t_{164} = \max\{t_{165}\} + 0.5 = 1.5$
 $t_{165} = \max\{t_{166}\} + 0.5 = 1.5$
 $t_{166} = \max\{t_{167}\} + 0.5 = 1.5$
 $t_{167} = \max\{t_{168}\} + 0.5 = 1.5$
 $t_{168} = \max\{t_{169}\} + 0.5 = 1.5$
 $t_{169} = \max\{t_{170}\} + 0.5 = 1.5$
 $t_{170} = \max\{t_{171}\} + 0.5 = 1.5$
 $t_{171} = \max\{t_{172}\} + 0.5 = 1.5$
 $t_{172} = \max\{t_{173}\} + 0.5 = 1.5$
 $t_{173} = \max\{t_{174}\} + 0.5 = 1.5$
 $t_{174} = \max\{t_{175}\} + 0.5 = 1.5$
 $t_{175} = \max\{t_{176}\} + 0.5 = 1.5$
 $t_{176} = \max\{t_{177}\} + 0.5 = 1.5$
 $t_{177} = \max\{t_{178}\} + 0.5 = 1.5$
 $t_{178} = \max\{t_{179}\} + 0.5 = 1.5$
 $t_{179} = \max\{t_{180}\} + 0.5 = 1.5$
 $t_{180} = \max\{t_{181}\} + 0.5 = 1.5$
 $t_{181} = \max\{t_{182}\} + 0.5 = 1.5$
 $t_{182} = \max\{t_{183}\} + 0.5 = 1.5$
 $t_{183} = \max\{t_{184}\} + 0.5 = 1.5$
 $t_{184} = \max\{t_{185}\} + 0.5 = 1.5$
 $t_{185} = \max\{t_{186}\} + 0.5 = 1.5$
 $t_{186} = \max\{t_{187}\} + 0.5 = 1.5$
 $t_{187} = \max\{t_{188}\} + 0.5 = 1.5$
 $t_{188} = \max\{t_{189}\} + 0.5 = 1.5$
 $t_{189} = \max\{t_{190}\} + 0.5 = 1.5$
 $t_{190} = \max\{t_{191}\} + 0.5 = 1.5$
 $t_{191} = \max\{t_{192}\} + 0.5 = 1.5$
 $t_{192} = \max\{t_{193}\} + 0.5 = 1.5$
 $t_{193} = \max\{t_{194}\} + 0.5 = 1.5$
 $t_{194} = \max\{t_{195}\} + 0.5 = 1.5$
 $t_{195} = \max\{t_{196}\} + 0.5 = 1.5$
 $t_{196} = \max\{t_{197}\} + 0.5 = 1.5$
 $t_{197} = \max\{t_{198}\} + 0.5 = 1.5$
 $t_{198} = \max\{t_{199}\} + 0.5 = 1.5$
 $t_{199} = \max\{t_{200}\} + 0.5 = 1.5$
 $t_{200} = \max\{t_{201}\} + 0.5 = 1.5$
 $t_{201} = \max\{t_{202}\} + 0.5 = 1.5$
 $t_{202} = \max\{t_{203}\} + 0.5 = 1.5$
 $t_{203} = \max\{t_{204}\} + 0.5 = 1.5$
 $t_{204} = \max\{t_{205}\} + 0.5 = 1.5$
 $t_{205} = \max\{t_{206}\} + 0.5 = 1.5$
 $t_{206} = \max\{t_{207}\} + 0.5 = 1.5$
 $t_{207} = \max\{t_{208}\} + 0.5 = 1.5$
 $t_{208} = \max\{t_{209}\} + 0.5 = 1.5$
 $t_{209} = \max\{t_{210}\} + 0.5 = 1.5$
 $t_{210} = \max\{t_{211}\} + 0.5 = 1.5$
 $t_{211} = \max\{t_{212}\} + 0.5 = 1.5$
 $t_{212} = \max\{t_{213}\} + 0.5 = 1.5$
 $t_{213} = \max\{t_{214}\} + 0.5 = 1.5$
 $t_{214} = \max\{t_{215}\} + 0.5 = 1.5$
 $t_{215} = \max\{t_{216}\} + 0.5 = 1.5$
 $t_{216} = \max\{t_{217}\} + 0.5 = 1.5$
 $t_{217} = \max\{t_{218}\} + 0.5 = 1.5$
 $t_{218} = \max\{t_{219}\} + 0.5 = 1.5$
 $t_{219} = \max\{t_{220}\} + 0.5 = 1.5$
 $t_{220} = \max\{t_{221}\} + 0.5 = 1.5$
 $t_{221} = \max\{t_{222}\} + 0.5 = 1.5$
 $t_{222} = \max\{t_{223}\} + 0.5 = 1.5$
 $t_{223} = \max\{t_{224}\} + 0.5 = 1.5$
 $t_{224} = \max\{t_{225}\} + 0.5 = 1.5$
 $t_{225} = \max\{t_{226}\} + 0.5 = 1.5$
 $t_{226} = \max\{t_{227}\} + 0.5 = 1.5$
 $t_{227} = \max\{t_{228}\} + 0.5 = 1.5$
 $t_{228} = \max\{t_{229}\} + 0.5 = 1.5$
 $t_{229} = \max\{t_{230}\} + 0.5 = 1.5$
 $t_{230} = \max\{t_{231}\} + 0.5 = 1.5$
 $t_{231} = \max\{t_{232}\} + 0.5 = 1.5$
 $t_{232} = \max\{t_{233}\} + 0.5 = 1.5$
 $t_{233} = \max\{t_{234}\} + 0.5 = 1.5$
 $t_{234} = \max\{t_{235}\} + 0.5 = 1.5$
 $t_{235} = \max\{t_{236}\} + 0.5 = 1.5$
 $t_{236} = \max\{t_{237}\} + 0.5 = 1.5$
 $t_{237} = \max\{t_{238}\} + 0.5 = 1.5$
 $t_{238} = \max\{t_{239}\} + 0.5 = 1.5$
 $t_{239} = \max\{t_{240}\} + 0.5 = 1.5$
 $t_{240} = \max\{t_{241}\} + 0.5 = 1.5$
 $t_{241} = \max\{t_{242}\} + 0.5 = 1.5$
 $t_{242} = \max\{t_{243}\} + 0.5 = 1.5$
 $t_{243} = \max\{t_{244}\} + 0.5 = 1.5$
 $t_{244} = \max\{t_{245}\} + 0.5 = 1.5$
 $t_{245} = \max\{t_{246}\} + 0.5 = 1.5$
 $t_{246} = \max\{t_{247}\} + 0.5 = 1.5$
 $t_{247} = \max\{t_{248}\} + 0.5 = 1.5$
 $t_{248} = \max\{t_{249}\} + 0.5 = 1.5$
 $t_{249} = \max\{t_{250}\} + 0.5 = 1.5$
 $t_{250} = \max\{t_{251}\} + 0.5 = 1.5$
 $t_{251} = \max\{t_{252}\} + 0.5 = 1.5$
 $t_{252} = \max\{t_{253}\} + 0.5 = 1.5$
 $t_{253} = \max\{t_{254}\} + 0.5 = 1.5$
 $t_{254} = \max\{t_{255}\} + 0.5 = 1.5$
 $t_{255} = \max\{t_{256}\} + 0.5 = 1.5$
 $t_{256} = \max\{t_{257}\} + 0.5 = 1.5$
 $t_{257} = \max\{t_{258}\} + 0.5 = 1.5$
 $t_{258} = \max\{t_{259}\} + 0.5 = 1.5$
 $t_{259} = \max\{t_{260}\} + 0.5 = 1.5$
 $t_{260} = \max\{t_{261}\} + 0.5 = 1.5$
 $t_{261} = \max\{t_{262}\} + 0.5 = 1.5$
 $t_{262} = \max\{t_{263}\} + 0.5 = 1.5$
 $t_{263} = \max\{t_{264}\} + 0.5 = 1.5$
 $t_{264} = \max\{t_{265}\} + 0.5 = 1.5$
 $t_{265} = \max\{t_{266}\} + 0.5 = 1.5$
 $t_{266} = \max\{t_{267}\} + 0.5 = 1.5$
 $t_{267} = \max\{t_{268}\} + 0.5 = 1.5$
 $t_{268} = \max\{t_{269}\} + 0.5 = 1.5$
 $t_{269} = \max\{t_{270}\} + 0.5 = 1.5$
 $t_{270} = \max\{t_{271}\} + 0.5 = 1.5$
 $t_{271} = \max\{t_{272}\} + 0.5 = 1.5$
 $t_{272} = \max\{t_{273}\} + 0.5 = 1.5$
 $t_{273} = \max\{t_{274}\} + 0.5 = 1.5$
 $t_{274} = \max\{t_{275}\} + 0.5 = 1.5$
 $t_{275} = \max\{t_{276}\} + 0.5 = 1.5$
 $t_{276} = \max\{t_{277}\} + 0.5 = 1.5$
 $t_{277} = \max\{t_{278}\} + 0.5 = 1.5$
 $t_{278} = \max\{t_{279}\} + 0.5 = 1.5$
 $t_{279} = \max\{t_{280}\} + 0.5 = 1.5$
 $t_{280} = \max\{t_{281}\} + 0.5 = 1.5$
 $t_{281} = \max\{t_{282}\} + 0.5 = 1.5$
 $t_{282} = \max\{t_{283}\} + 0.5 = 1.5$
 $t_{283} = \max\{t_{284}\} + 0.5 = 1.5$
 $t_{284} = \max\{t_{285}\} + 0.5 = 1.5$
 $t_{285} = \max\{t_{286}\} + 0.5 = 1.5$
 $t_{286} = \max\{t_{287}\} + 0.5 = 1.5$
 $t_{287} = \max\{t_{288}\} + 0.5 = 1.5$
 $t_{288} = \max\{t_{289}\} + 0.5 = 1.5$
 $t_{289} = \max\{t_{290}\} + 0.5 = 1.5$
 $t_{290} = \max\{t_{291}\} + 0.5 = 1.5$
 $t_{291} = \max\{t_{292}\} + 0.5 = 1.5$
 $t_{292} = \max\{t_{293}\} + 0.5 = 1.5$
 $t_{293} = \max\{t_{294}\} + 0.5 = 1.5$
 $t_{294} = \max\{t_{295}\} + 0.5 = 1.5$
 $t_{295} = \max\{t_{296}\} + 0.5 = 1.5$
 $t_{296} = \max\{t_{297}\} + 0.5 = 1.5$
 $t_{297} = \max\{t_{298}\} + 0.5 = 1.5$
 $t_{298} = \max\{t_{299}\} + 0.5 = 1.5$
 $t_{299} = \max\{t_{300}\} + 0.5 = 1.5$
 $t_{300} = \max\{t_{301}\} + 0.5 = 1.5$
 $t_{301} = \max\{t_{302}\} + 0.5 = 1.5$
 $t_{302} = \max\{t_{303}\} + 0.5 = 1.5$
 $t_{303} = \max\{t_{304}\} + 0.5 = 1.5$
 $t_{304} = \max\{t_{305}\} + 0.5 = 1.5$
 $t_{305} = \max\{t_{306}\} + 0.5 = 1.5$
 $t_{306} = \max\{t_{307}\} + 0.5 = 1.5$
 $t_{307} = \max\{t_{308}\} + 0.5 = 1.5$
 $t_{308} = \max\{t_{309}\} + 0.5 = 1.5$
 $t_{309} = \max\{t_{310}\} + 0.5 = 1.5$
 $t_{310} = \max\{t_{311}\} + 0.5 = 1.5$
 $t_{311} = \max\{t_{312}\} + 0.5 = 1.5$
 $t_{312} = \max\{t_{313}\} + 0.5 = 1.5$
 $t_{313} = \max\{t_{314}\} + 0.5 = 1.5$
 $t_{314} = \max\{t_{315}\} + 0.5 = 1.5$
 $t_{315} = \max\{t_{316}\} + 0.5 = 1.5$
 $t_{316} = \max\{t_{317}\} + 0.5 = 1.5$
 $t_{317} = \max\{t_{318}\} + 0.5 = 1.5$
 $t_{318} = \max\{t_{319}\} + 0.5 = 1.5$
 $t_{319} = \max\{t_{320}\} + 0.5 = 1.5$
 $t_{320} = \max\{t_{321}\} + 0.5 = 1.5$
 $t_{321} = \max\{t_{322}\} + 0.5 = 1.5$
 $t_{322} = \max\{t_{323}\} + 0.5 = 1.5$
 $t_{323} = \max\{t_{324}\} + 0.5 = 1.5$
 $t_{324} = \max\{t_{325}\} + 0.5 = 1.5$
 $t_{325} = \max\{t_{326}\} + 0.5 = 1.5$
 $t_{326} = \max\{t_{327}\} + 0.5 = 1.5$
 $t_{327} = \max\{t_{328}\} + 0.5 = 1.5$
 $t_{328} = \max\{t_{329}\} + 0.5 = 1.5$
 $t_{329} = \max\{t_{330}\} + 0.5 = 1.5$
 $t_{330} = \max\{t_{331}\} + 0.5 = 1.5$
 $t_{331} = \max\{t_{332}\} + 0.5 = 1.5$
 $t_{332} = \max\{t_{333}\} + 0.5 = 1.5$